

AUTO.MPG DATA AUDIT AND EXPLORATORY DATA ANALYSIS (EDA)

1. Introduction

The Auto MPG dataset tracks 398 vehicles (1970–1982). The 1973 oil crisis and government regulations forced automakers to shift focus from heavy muscle cars to fuel efficiency (MPG). This report audits the data, identifies outliers, maps variable relationships, and checks readiness for predictive modeling.

2. Data Audit

The dataset has 398 rows and 9 columns. The horsepower column was missing 6 values, which were filled with the column median (93.5 hp) because the data are right-skewed; an average would be pulled up by a few rare, high-powered sports cars.

For categorical data, origin numbers (1, 2, 3) were mapped to 'USA', 'Europe', and 'Japan'. The raw car names were split to extract the manufacturer's brand (the first word). This process exposed text typos (e.g., *chevroelt* for Chevrolet, *toyouta* for Toyota), which were corrected. Ford (51) and Chevrolet (47) are the top brands.

3. Methodology

The analysis followed four direct steps:

- **Summary Statistics:** Calculated basic metrics (mean, median, min, max) to evaluate distributions.
- **Outlier Detection:** Used the Interquartile Range (IQR) method, flagging values sitting more than $1.5 \times \text{IQR}$ past the middle 50% of the data.
- **Scatter Plots:** Plotted weight against MPG. A robust regression trendline was used to keep outliers from warping the line of best fit.
- **Correlation Matrix:** Generated a Pearson correlation grid to calculate how variables overlap and check for multicollinearity.

4. Visualization

Two charts were generated in Python to visually inspect the dataset's structure.

Figure 1: Horsepower Spread and Outliers by Region

Figure 1 uses boxplots to map horsepower across regions. It shows that American cars occupied a heavy engineering space, with a wide spread reaching up to 230 hp. European and Japanese cars are tightly compressed below 125 hp. The dots floating above the European and Japanese whiskers are the exact outliers caught by our math script.

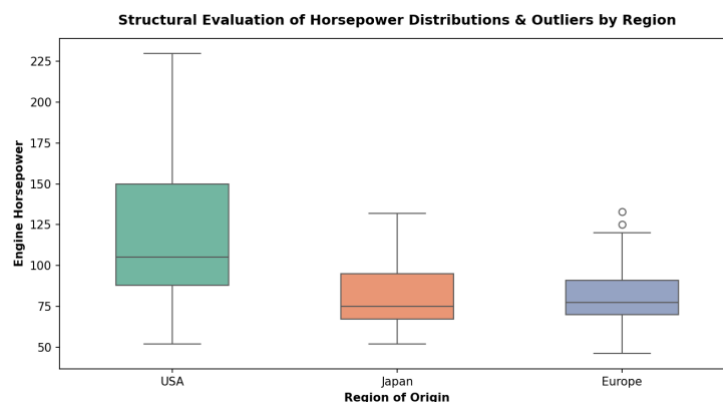


Figure 2: Car Weight vs. Fuel Efficiency (MPG)

Figure 2 shows vehicle weight versus MPG. The dots show a clear downward curve, proving that lighter cars get better mileage. The red robust trendline cuts through the center of the data. Because it down-weights outliers, it confirms that every extra pound added to a car consistently cuts its fuel efficiency.

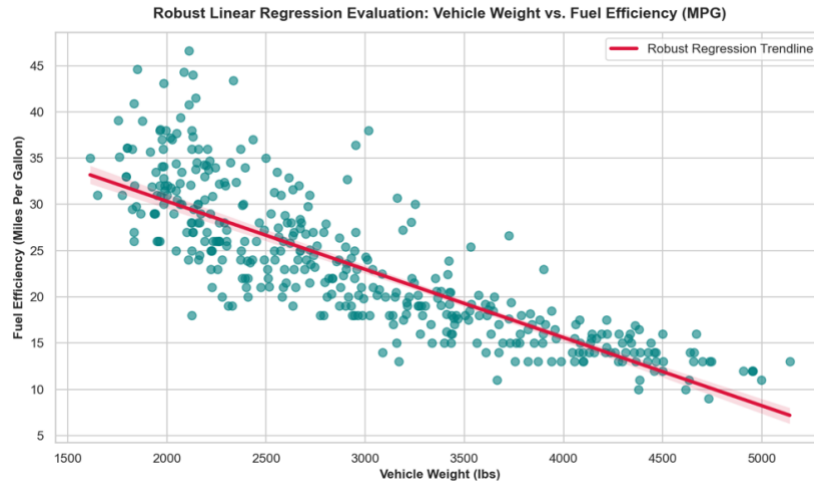
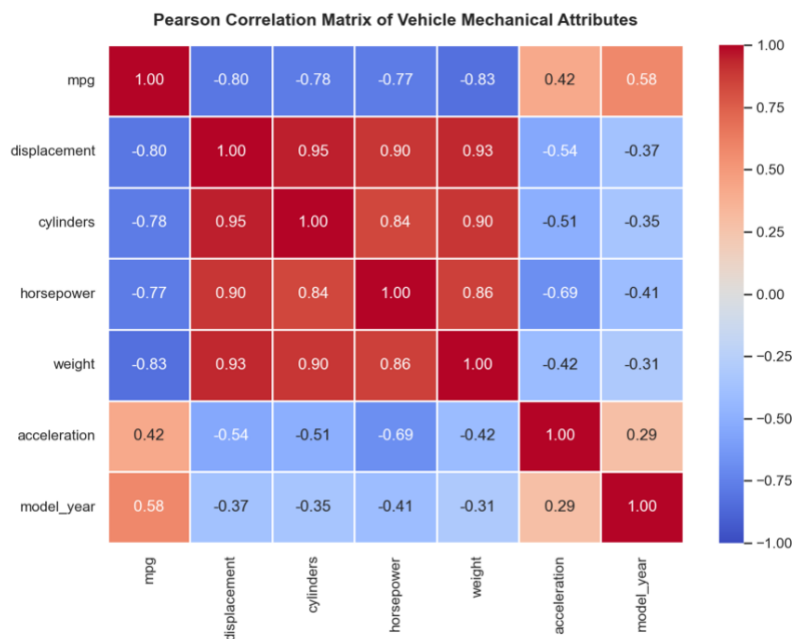


Figure 3: Pearson Correlation Matrix of Vehicle Mechanical Attributes

Figure 3 presents a correlation heatmap across all numeric variables in the dataset. Red squares indicate strong positive correlations, while blue squares indicate strong negative relationships. The matrix visually highlights how tightly clustered the engineering features are, serving as a diagnostic warning for future modeling phases.



5. Key Insight

Our outlier scan flagged 11 cars with extreme horsepower but found zero outliers for weight or engine size (displacement). These 11 outliers are all heavy American V8 models, such as the *Buick Estate Wagon* and *Plymouth Fury*. While American assembly lines built standard-sized frames, they installed massive, inefficient engines. This placed domestic brands in the lowest-efficiency zone of our scatter plot, while foreign makers optimized for lightweight 4-cylinder commuter cars.

The most critical warning for modeling came from the correlation grid. The analysis revealed severe multicollinearity (overlapping data) among weight, displacement, and horsepower, with correlation coefficients exceeding 0.85. These features measure the same underlying trait: larger engines require heavier frames and generate more power. Because weight has the strongest negative association with MPG ($r = -0.83$), mass is the primary driver of poor mileage. Using weight, engine size, and horsepower together in a basic multiple regression model will break the math and render the results unreliable.

6. Conclusions

The dataset is clean and ready for modeling. To build a successful model, next, three steps are required:

- **Log Transformations:** Apply a log transformation ($\log(y)$) to MPG and engine size to flatten the curved relationship and satisfy regression rules.
- **Feature Selection:** Due to severe math overlap (multicollinearity) between weight, engine size, and horsepower, we cannot use all three. We must drop redundant features or use a regularized model (like Ridge/Lasso regression).
- **Categorical Encoding:** Convert the origin and brand text columns into binary dummy variables (One-Hot Encoding) before modeling.

Note: Generative AI was used as an auxiliary tool for debugging Python code blocks, organizing data-cleaning steps, and refining the report's language for clarity.